

Stress Testing — Outpatient Evaluation for Obstructive CAD

Non-acute chest pain in clinic asks one question: **is this obstructive (>50%) CAD?** — distinct from the ED's "is this ACS?" Risk-stratify by pretest probability, then test. Adapted from TeachIM.org (Fainstad; Dec 2020), CC BY-NC 4.0 · 2021 AHA/ACC Chest Pain + 2023 CCD updates layered.

1 · Pretest probability → action

CAD Consortium Clinical Score (age, sex, pain character, DM/HTN/dyslipidemia/smoking). A check on gestalt, not a replacement.

Band	Action
Low <5%	Seek alternative cause; no stress test. CAC = 0 supports deferral.
Intermediate 5–70%	Stress test to confirm/exclude obstructive CAD.
High >70%	ECG + imaging to exclude high-grade dz; start empiric medical rx.

High-risk testing goal = exclude **left main, proximal LAD, 3-vessel disease, EF<35%** (sudden-death risk), not to diagnose CAD.

2 · Choosing the stress test

Branch	Yes → / No →
Can exercise?	Yes → exercise test (functional info, reproduces symptom) · No → pharmacologic
ECG interpretable?*	Yes → exercise treadmill ECG (cheapest) · No → imaging stress
Echo vs MPI/ SPECT	Institutional; both give EF + ischemia. Echo adds valves; nuclear for poor windows.

*Uninterpretable ECG = LBBB, paced rhythm, pre-excitation (WPW), LVH w/ strain, or >1 mm resting ST-dep. In LBBB prefer **vasodilator** MPI.

3 · "Markedly positive" → define anatomy

(CCTA vs left heart cath) — otherwise positive test → medical therapy first.

- **ECG / exercise:** ST-dep at low workload · ST-elevation · exertional hypotension.
- **Imaging:** rest EF <35% · EF drop or LV dilation with stress · ischemia >2 territories.

4 · Empiric medical therapy (stable CAD)

Bucket	Therapy
Risk-factor	Diet/exercise, smoking cessation, A1c <7.0, BP <130/80
Cardioprotective	ASA 81 · high-intensity statin (± ezetimibe/PCSK9) · ACE-I/ARB if DM, CKD, EF<40%
Anti-anginal	SL nitroglycerin PRN · beta-blocker · long-acting nitrate or CCB · consider ranolazine

2023 CCD update: beta-blocker not mandatory without prior MI/EF≤50%; add SGLT2i or GLP-1 RA if diabetes (SGLT2i if HF); colchicine 0.5 mg daily reasonable (2a, LoDoCo2).

5 · 2021 Chest Pain Guideline

- Describe pain as **cardiac / possibly cardiac / noncardiac** — retire "typical/atypical."
- **CAC score** = useful first-line gatekeeper in low-risk stable patients (CAC 0 → low risk).
- **CCTA = Class 1 first-line** anatomic test for intermediate–high risk (favored if <65 y / less obstructive dz); stress imaging favored if >65 y / more obstructive dz.

6 · Treat medically first (ISCHEMIA)

In stable CAD with moderate–severe ischemia, an initial invasive strategy did **not** reduce death/MI vs. optimal medical therapy; revascularization improves angina quality of life. Reserve cath for refractory symptoms or high-risk anatomy.

Do / Avoid

- **Do:** name the question, score the pretest probability, match the test to exercise ability + ECG interpretability, treat medically before cath.
- **Avoid:** reflex cath for moderate ischemia; exercise ECG when the baseline ECG is uninterpretable; stress testing a low-pretest patient.
- **Switch questions** if dynamic ECG changes / positive troponin / crescendo pattern → acute ACS pathway, not outpatient stress test.